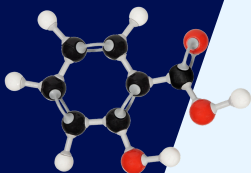
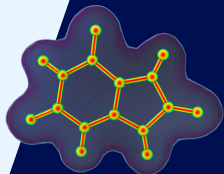


ACETONE

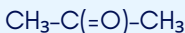
Simple Molecule with Major
Industrial Uses

Compound Identity

Common name: Acetone
IUPAC name: Propan-2-one
Molecular formula: C_3H_6O
Molar mass: 58.08 g/mol



Structural formula



Major functional group:

- Ketone ($C=O$)

Important Properties

- Colorless liquid
- Highly volatile; evaporates easily
- Miscible with water
- Flammable
- Polar aprotic solvent



What Is Amazing About It?

Acetone is a small molecule but an extremely useful solvent. Its polarity allows it to dissolve many organic substances, while its volatility allows it to evaporate readily after use.



Uses and Value

- Found in products such as some paint and nail-polish removers
- Used in manufacturing plastics, fibers, pharmaceuticals, and other chemicals
- Valuable as an industrial solvent and chemical intermediate



Risks and Safety

- Highly flammable
- Avoid ignition sources and unnecessary inhalation of concentrated vapors.
- Use appropriate ventilation and laboratory safety procedures.



Chemical Engineering Connection

Acetone is strongly connected to separation processes, especially distillation. Because acetone is volatile, engineers can use differences in volatility to separate it from less-volatile components.



References

PubChem. (n.d.). Acetone. PubChem. <https://pubchem.ncbi.nlm.nih.gov/compound/Acetone>
Informatics, N. O. O. D. A. (n.d.). Acetone. <https://webbook.nist.gov/cgi/cbook.cgi?ID=C67641>

